(KTXFI1EBNF) Physics I. Lecture

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April 27, 2026

Current Updates and Announcements

- April 27, May 4, and go...
- Homework No. 2 is now available. The deadline is May 4, 2026 at midnight.
- Homework No. 3 will be available tonight. The deadline will be May 10, 2026 at midnight.

Where are we? I

1. **Mechanics I.:** Newton's laws, equation of motion (review). Momentum. General concept of mechanical work, work theorem. Forms of mechanical energy.
2. **Mechanics II.:** Weight force, gravitational force, gravitational field. Work of the gravitational force. Conservative force field. Dissipative force field. Mechanical power.
3. **Mechanics of Rigid Bodies I.** Concept of a rigid body. Forces acting on a rigid body. Motion of a rigid body. Concept of the centre of mass. Location of the centre of mass. Torque. Angular momentum. Energy of a rotating rigid body, angular momentum of a rotating rigid body, Steiner's theorem.
4. **Mechanics of Rigid Bodies II.** Principal theorems of rigid body motion; conservation laws. Collisions.
5. **Oscillations I.** Dynamic condition of harmonic oscillatory motion. Differential equation of harmonic undamped oscillatory motion.

Where are we? II

6. **Oscillations II.** Superposition of oscillations. Decomposition of oscillations into components. Damped oscillations. Forced oscillations, resonance. Coupled oscillations.
7. **Wave phenomena:** Reflection of waves, refraction of waves, diffraction of waves. Huygens–Fresnel principle. Interference of waves. Polarization of waves.
8. **Elements of acoustics:** Basics of acoustics. Fundamental concepts. Doppler effect. Sound intensity, loudness according to Weber–Fechner's law.
9. **Thermodynamics I:** Temperature scales, thermal expansion, thermometers. State variables, ideal gas, state changes, ideal gas equation of state, Boyle's law, Gay-Lussac's laws, universal (combined) gas law, heat, specific heat, molar heat, heat capacity.
10. **Thermodynamics II:** Expansion work, internal energy, the first law of thermodynamics, forms of the first law in different state changes.

Where are we? III

11. **Thermodynamics III:** Cyclic processes, Carnot cycle, Carnot's theorem, reduced heat, entropy, the second law of thermodynamics.
12. **Thermodynamics IV:** Kinetic theory of gases, molecular thermodynamics. Distributions, distribution functions. Elementary statistics.
13. **Electrostatics I.:** Electric charge: Fundamental electrical phenomena, electric charge and matter, Coulomb's law (force between two point charges, unit of electric charge, permittivity of vacuum).
14. **Electrostatics II.:** The electric field: The electrostatic field (electric field strength, electric field lines). Determination of electric field strength. Gauss's law (flux of the electric field). Electric field of a point charge.

Where are we? IV

15. **Electrostatics III.:** Electrostatic potential: Work of the electric field (work done in an electric field). Electric potential (electrostatic potential). Electric potential difference, voltage. Electric potential of charge systems at small and large distances. Electrostatic energy.
16. **Elements of physical or wave optics.** *Branches of optics. Light interference, light diffraction, diffraction through a slit, optical grating, light polarization. Light as an electromagnetic wave: Poynting vector. Optical instruments.*

Elements of physical and wave optics I

- Branches of optics: When each model is useful?

Branch	Main idea	Typical phenomena/instruments
Geometrical optics	Light as rays; wavelength is negligible	Lenses, mirrors, cameras, microscopes
Wave optics	Light as a wave; phase matters	Interference, diffraction, coherence

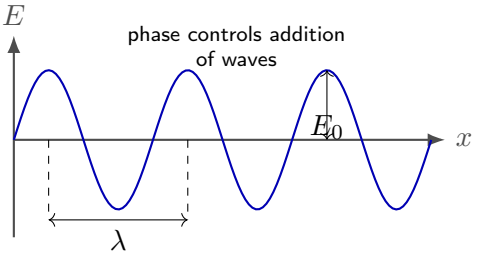
- **Rule of thumb:** Ray optics works when object sizes and apertures are much larger than the wavelength. Wave effects dominate when path differences or apertures are comparable to λ.

Elements of physical and wave optics II

- **Light as a wave: phase, amplitude, and intensity**

- A monochromatic plane wave can be written as

$$
\begin{aligned}
E(x,t) &= E_0 \cos(kx - \omega t + \phi), \\
k &= \frac{2\pi}{\lambda}, \\
\omega &= 2\pi f.
\end{aligned}
$$



- In linear optical media, measured brightness is proportional to the time averaged square of the field:

$$I \sim \left\langle E^2 \right\rangle_t .$$

Elements of physical and wave optics III

- **Two-source interference: path difference decides brightness**

- Two coherent waves arriving at a point combine by superposition.

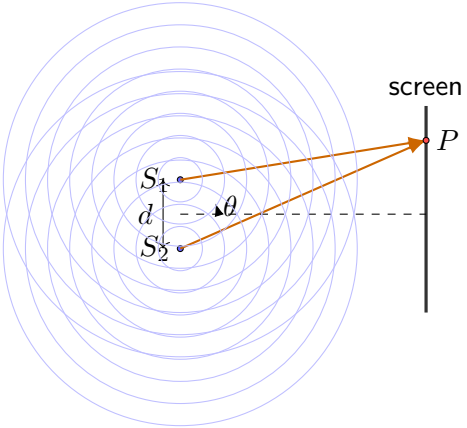
$$\delta = d\sin\theta, \qquad \Delta\phi = \frac{2\pi}{\lambda}\delta.$$

- Constructive interference:

$$\delta = m\lambda \quad (m = 0, \pm 1, \pm 2, \ldots).$$

- Destructive interference:

$$\delta = \left(m + \frac{1}{2}\right)\lambda.$$



Elements of physical and wave optics IV

- **Interference intensity for equal amplitudes**

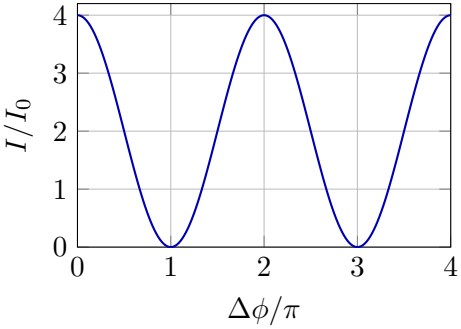
- For two waves of equal amplitude,

$$E = E_0 \cos \omega t + E_0 \cos(\omega t + \Delta\phi),$$

so the intensity is

$$I = 4I_0 \cos^2 \left(\frac{\Delta\phi}{2} \right).$$

- The observable pattern is a map of phase differences.



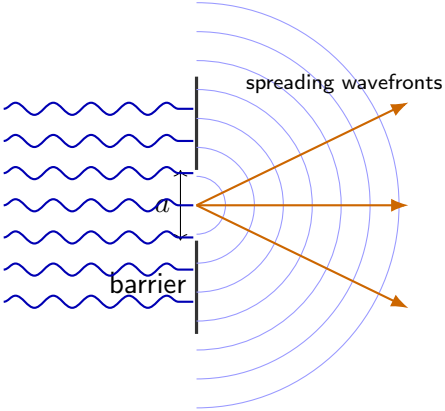
Elements of physical and wave optics V

- Diffraction: every aperture becomes a source of wavelets

- **Huygens–Fresnel principle:** each point of a wavefront acts as a source of secondary spherical wavelets.
- Diffraction is strongest when

$$a \sim \lambda,$$

  where a is the aperture or obstacle size.
- Wave optics explains why perfectly sharp shadows and unlimited resolution are impossible.



Elements of physical and wave optics VI

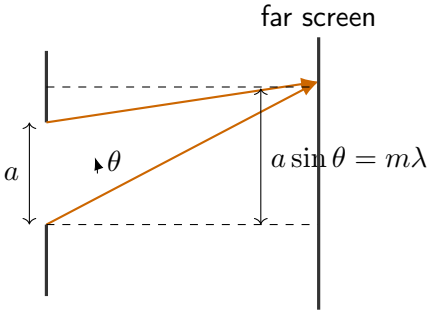
- Fraunhofer diffraction: far-field geometry

- Fraunhofer diffraction describes the far field, or the focal plane of a lens.
- For a slit of width a, rays from different parts of the slit interfere at angle θ.

$$\Delta = a \sin\theta.$$

- The minima occur when the aperture can be divided into canceling pairs:

$$a \sin\theta = m\lambda, \qquad m = \pm 1, \pm 2, \ldots$$



Elements of physical and wave optics VII

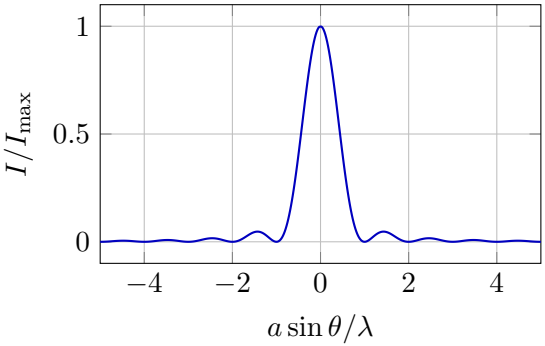
- **Single-slit diffraction pattern**

- For a uniformly illuminated slit,

$$I(\theta) = I_{\max}\left(\frac{\sin\beta}{\beta}\right)^2,$$
$$\beta = \frac{\pi a \sin\theta}{\lambda}.$$

Important features:

- central maximum is the widest and brightest;
- dark fringes: $a\sin\theta = m\lambda$;
- increasing a narrows the pattern.



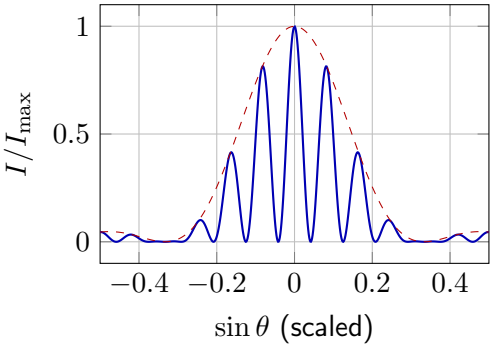
Elements of physical and wave optics VIII

- **Double slit: interference modulated by diffraction**

- Real double slits have finite slit width a and separation d.

$$I(\theta) = I_{\max} \cos^2\left(\frac{\pi d \sin\theta}{\lambda}\right)\left(\frac{\sin\beta}{\beta}\right)^2.$$

- Interference fringes sit inside the single-slit diffraction envelope.



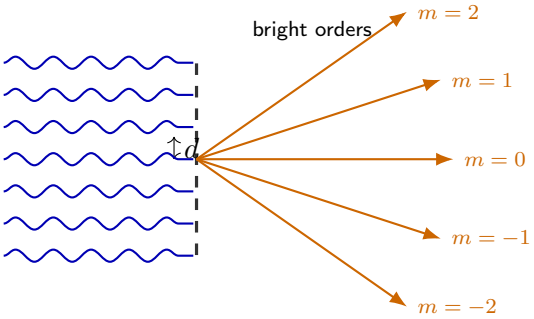
Elements of physical and wave optics IX

- **Optical grating: many slits sharpen the maxima**

- For a grating with spacing d, principal maxima satisfy

$$\begin{aligned} d\sin\theta &= m\lambda, \\ m &= 0, \pm 1, \pm 2, \ldots \end{aligned}$$

- Because many slits interfere, the maxima become narrow and intense. This makes gratings useful for spectroscopy.



Elements of physical and wave optics X

- **Grating dispersion and resolving power**

- Different wavelengths leave at different angles:

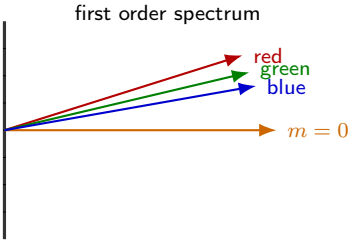
$$d \sin\theta = m\lambda.$$

- Angular dispersion:

$$\frac{d\theta}{d\lambda} = \frac{m}{d\cos\theta}.$$

- Resolving power for N illuminated grooves:

$$R = \frac{\lambda}{\Delta\lambda} = mN.$$

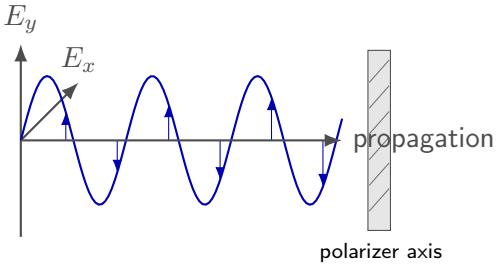


Elements of physical and wave optics XI

- **Polarization: direction of the electric field**

- Polarization describes how the transverse electric field oscillates.
  - Linear: fixed direction.
  - Circular: constant magnitude, rotating direction.
  - Elliptical: general case.
- For ideal polarizer axes separated by angle α,

$$I = I_0 \cos^2 \alpha \quad \text{(Malus' law).}$$

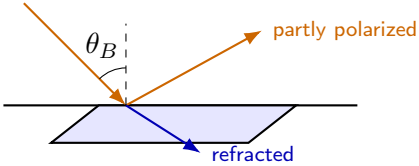


Elements of physical and wave optics XII

- **How polarization is produced and used**

- Common mechanisms:
  - selective absorption by polarizing filters;
  - reflection near Brewster's angle;
  - birefringence in anisotropic crystals;
  - scattering by small particles.

  Applications include LCDs, stress analysis, sunglasses, polarimetry, and astronomy.



Brewster reflection

Electromagnetic waves I

- **Maxwell equations in vacuum:** In a source-free vacuum,

$$
\begin{aligned}
\nabla \cdot \mathbf{E} &= 0, \\
\nabla \cdot \mathbf{B} &= 0, \\
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, \\
\nabla \times \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.
\end{aligned}
$$

- These equations imply that changing electric fields create magnetic fields, and changing magnetic fields create electric fields. A self-sustaining electromagnetic wave can propagate through empty space.
- **Speed of light:** $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$.

Electromagnetic waves II

- **Derivation of the EM wave equation for $\mathbf{E}$**
  - Take the curl of Faraday's law:

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t}(\nabla \times \mathbf{B}).$$

  - Use the vector identity

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E}.$$

  - In vacuum, $\nabla \cdot \mathbf{E} = 0$, so

$$-\nabla^2 \mathbf{E} = -\frac{\partial}{\partial t}\left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\right).$$

Electromagnetic waves III

- Therefore

$$\boxed{\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}} \qquad \Longleftrightarrow \qquad \boxed{\nabla^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0.}$$

Electromagnetic waves IV

- **Derivation of the EM wave equation for $\mathbf{B}$**
  - Take the curl of Ampere–Maxwell's law:

$$\nabla \times (\nabla \times \mathbf{B}) = \mu_0 \epsilon_0 \frac{\partial}{\partial t}(\nabla \times \mathbf{E}).$$

  - Use $\nabla \cdot \mathbf{B} = 0$ and Faraday's law:

$$-\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial}{\partial t}\left(-\frac{\partial \mathbf{B}}{\partial t}\right).$$

  - Thus

$$\boxed{\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}} \quad \Longleftrightarrow \quad \boxed{\nabla^2 \mathbf{B} - \frac{1}{c^2}\frac{\partial^2 \mathbf{B}}{\partial t^2} = 0.}$$

Electromagnetic waves V

- Plane-wave solutions satisfy

$$\mathbf{E} \perp \mathbf{B} \perp \mathbf{k}, \qquad B_0 = \frac{E_0}{c}.$$

Electromagnetic waves VI

- **Relation Between the Electric and Magnetic Fields**
  - Assume similarly that
$$\mathbf{B}(\mathbf{r},t) = \mathbf{B}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}.$$
  - Using Faraday's law,
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t},$$
    we obtain
$$i\mathbf{k} \times \mathbf{E} = i\omega \mathbf{B}.$$
  - Hence
$$\boxed{\mathbf{B} = \frac{1}{\omega}\mathbf{k} \times \mathbf{E}}$$

Electromagnetic waves VII

and, since $\omega = ck$,

$$\boxed{B_0 = \frac{E_0}{c}} \qquad \text{or} \qquad \boxed{\frac{E_0}{B_0} = c.}$$

- The plane electromagnetic wave is transverse:

$$\boxed{\mathbf{k} \perp \mathbf{E}, \qquad \mathbf{k} \perp \mathbf{B}, \qquad \mathbf{E} \perp \mathbf{B}.}$$

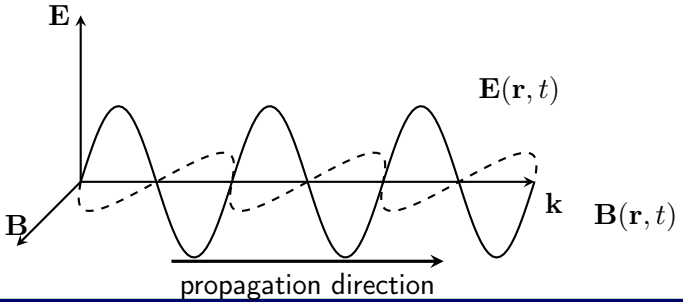
Electromagnetic waves VIII

- **Plane Electromagnetic Wave:** A real physical plane-wave solution may be written as

$$\mathbf{E}(\mathbf{r},t) = \mathbf{E}_0 \cos(\mathbf{k}\cdot\mathbf{r} - \omega t),$$

and

$$\mathbf{B}(\mathbf{r},t) = \frac{1}{\omega}\mathbf{k}\times\mathbf{E}_0 \cos(\mathbf{k}\cdot\mathbf{r} - \omega t).$$

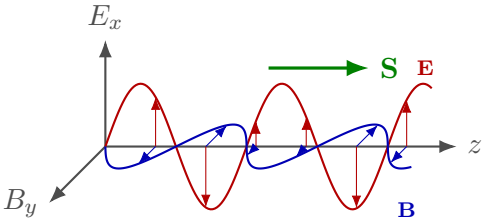


Electromagnetic waves IX

- **Plane electromagnetic wave and the Poynting vector**

- For a plane wave moving in $+z$,

$$\begin{aligned}\mathbf{E} &= E_0 \cos(kz - \omega t)\,\hat{\mathbf{x}},\\ \mathbf{B} &= B_0 \cos(kz - \omega t)\,\hat{\mathbf{y}}.\end{aligned}$$



- The energy flux density is the Poynting vector:

$$\boxed{\mathbf{S} = \frac{1}{\mu_0}\mathbf{E} \times \mathbf{B}.}$$

- For a plane wave,

$$\langle S \rangle = \frac{1}{2}\epsilon_0 c E_0^2.$$

Electromagnetic waves X

- **Energy density and radiation pressure**
  - The electromagnetic energy density is

$$u = \frac{1}{2}\epsilon_0 E^2 + \frac{1}{2\mu_0} B^2.$$

  - For a plane wave the electric and magnetic contributions are equal, so

$$u = \epsilon_0 E^2 = \frac{B^2}{\mu_0}.$$

  - The Poynting vector gives energy crossing unit area per unit time:

$$\mathbf{S} = u\, c\, \hat{\mathbf{k}} \quad \text{for a plane wave.}$$

Electromagnetic waves XI

- Radiation pressure follows from momentum transport:

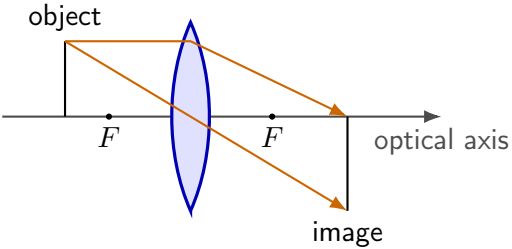
$$p_{\text{absorber}} = \frac{I}{c}, \qquad p_{\text{mirror}} = \frac{2I}{c}.$$

Optical instruments: lenses and mirrors I

- Optical instruments form images by controlling wavefronts.
  - Lenses refract light.
  - Mirrors reflect light.
  - Stops and apertures control brightness and resolution.
  - Detector or eye records the final image.

For thin lenses and spherical mirrors:

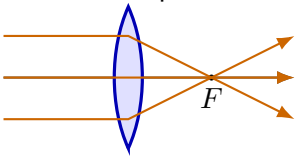
$$\frac{1}{f} = \frac{1}{s_o} + \frac{1}{s_i}, \qquad m = -\frac{s_i}{s_o}.$$



Optical instruments: lenses and mirrors II

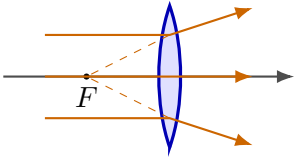
- **Convex and concave lenses**

Convex lens: converging. Parallel rays focus at the real focal point.



$f > 0$

Concave lens: diverging. Parallel rays appear to come from a virtual focal point.

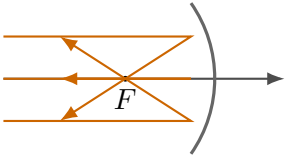


$f < 0$

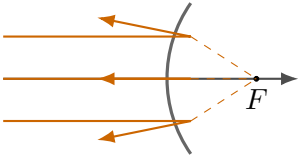
Optical instruments: lenses and mirrors III

- **Convex and concave spherical mirrors**

Concave spherical mirror: converging; real focus in front of mirror.



Convex spherical mirror: diverging; virtual focus behind mirror.



Convex mirrors give a wide field of view but virtual, reduced images.

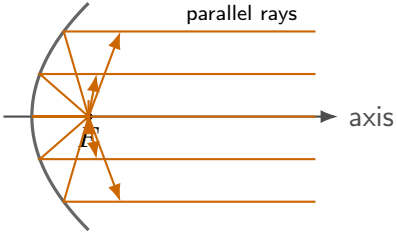
Optical instruments: lenses and mirrors IV

- **Concave parabolic mirror**

- A parabolic mirror focuses parallel incoming rays to one point without spherical aberration:

$$y^2 = 4fx.$$

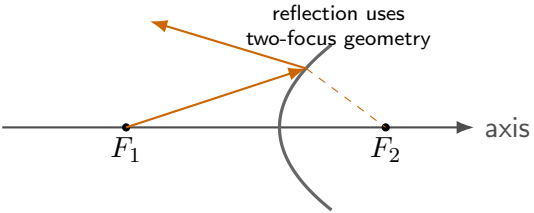
- It is ideal for on-axis astronomical telescopes and searchlights.



Optical instruments: lenses and mirrors V

- **Concave hyperbolic mirror**

- A hyperbolic mirror has two foci. A ray directed toward one focus reflects as if it came from the other focus.
- This property is used in Cassegrain and Ritchey–Chretien telescopes, where a convex hyperbolic secondary redirects light through the primary mirror.



Optical instruments: telescopes I

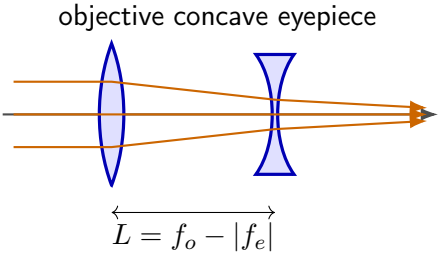
- **Galilean telescope: upright image, short tube**

- **Objective:** convex lens.

- **Eyepiece:** concave lens placed before the real focal point of the objective.
- For relaxed eye viewing, approximately:

$$L = f_o - |f_e|, \qquad M = -\frac{f_o}{f_e} > 0.$$

- It gives an upright image, but the field of view is narrow.



Optical instruments: telescopes II

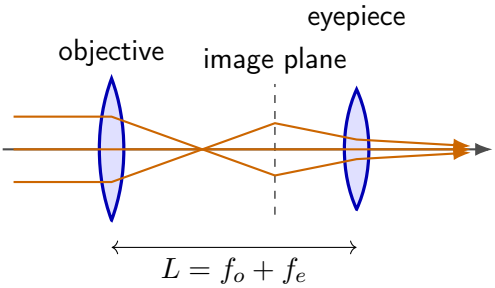
- **Keplerian telescope: real intermediate image**

- **Objective:** convex lens.

- **Eyepiece:** convex lens.
- For relaxed viewing:

$$L = f_o + f_e, \qquad M = -\frac{f_o}{f_e}.$$

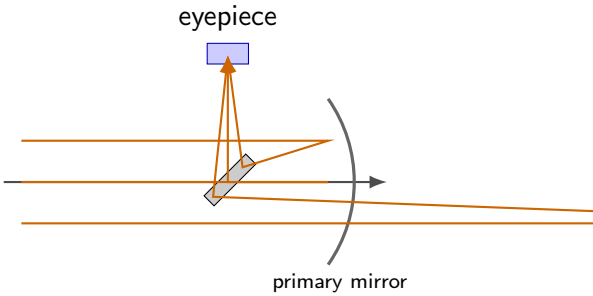
- The image is inverted, but the field of view is wider and the intermediate image plane can hold a reticle or detector.



Optical instruments: telescopes III

- **Newtonian reflector**

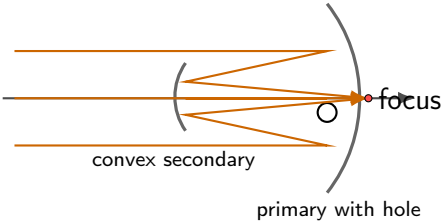
- A concave primary mirror forms a focus. A flat secondary mirror redirects the beam sideways to an eyepiece.
- **Strengths:** simple, economical, no chromatic aberration.
- **Limits:** off-axis coma and central obstruction.



Optical instruments: telescopes IV

- **Cassegrain reflector**

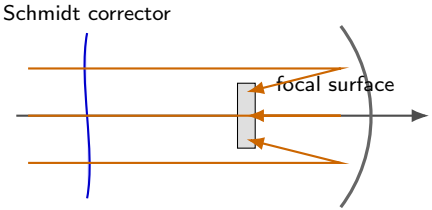
- A concave primary mirror and convex secondary mirror fold the optical path. The final focus lies behind the primary mirror.
- This gives a long effective focal length in a compact tube.



Optical instruments: telescopes V

- **Schmidt telescope / Schmidt camera**

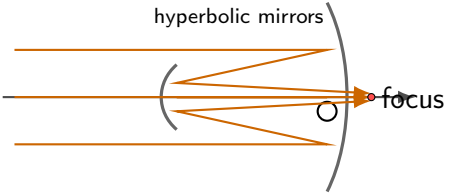
- A thin aspheric Schmidt corrector plate reduces spherical aberration from a spherical primary mirror.
- **Purpose:** wide-field imaging with a relatively simple primary mirror.



Optical instruments: telescopes VI

- **Ritchey–Chretien reflector**

- A Ritchey–Chretien telescope uses a concave hyperbolic primary and a convex hyperbolic secondary.
- **Advantage:** it corrects primary spherical aberration and coma, giving excellent wide-field astronomical imaging.
- **Trade-off:** curved focal plane and more difficult fabrication/alignment.



Optical instruments: telescopes VII

- **Lens telescopes versus mirror telescopes**

	Refracting telescopes	Reflecting telescopes
Main element	Objective lens	Primary mirror
Advantages	Sealed tube possible; stable alignment; high contrast; simple light path	No chromatic aberration; large apertures are practical; compact folded designs possible
Disadvantages	Chromatic aberration unless corrected; heavy large lenses; lens sag and absorption	Central obstruction; mirror coating aging; collimation and thermal effects
Best uses	Small high-quality visual telescopes, teaching, terrestrial viewing	Large astronomical telescopes, deep-sky imaging, professional observatories

Optical instruments: telescopes VIII

- **Final comparison:** Refractors are often mechanically simple and high contrast at small aperture. Reflectors dominate at large aperture because mirrors can be supported from behind and avoid chromatic aberration.

The End

Thank you for your attention!